

Exact and approximate methods for a one-dimensional minimax bin-packing problem

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Abstract One-dimensional bin-packing problems require the assignment of a collection of items to bins with the goal of optimizing some criterion related to the number of bins used or the ‘weights’ of the items assigned to the bins. In many instances, the number of bins is fixed and the goal is to assign the items such that the sums of the item weights for each bin are approximately equal. Among the possible applications of one-dimensional bin-packing in the field of psychology are the assignment of subjects to treatments and the allocation of students to groups. An especially important application in the psychometric literature pertains to splitting of a set of test items to create distinct subtests, each containing the same number of items, such that the maximum sum of item weights across all bins is minimized. In this context, the weights typically correspond to item statistics derived from difficulty and discrimination indices. We present a mixed zero-one integer linear programming (MZOILP) formulation of this one-dimensional minimax bin-packing problem and develop an approximate procedure for its solution that is based on the simulated annealing algorithm. In two comparisons that focused on 34 practically-sized test problems (up to 6000 items and 300 bins), the simulated annealing heuristic generally provided better solutions than were obtained when using a commercial mathematical programming software package to solve the MZOILP formulation directly.

Keywords One-dimensional bin-packing · Combinatorial optimization · Integer programming · Simulated annealing · Test splitting

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1 Introduction

Bin-packing problems require the assignment of a collection of *items* to a set of *bins*. Each item is measured on one or more *dimensions* and the measurements are typically referred to as ‘*item weights*’ for the packing process. *One-dimensional* bin-packing problems have been well-studied in the operations research literature and several excellent surveys and theoretical analyses are available (Anily et al. 1984; Csirik et al. 2006; Coffman et al. 1997; Fleszar and Hindi 2002; Hall et al. 1988; Johnson et al. 1974). A classic example of the one-dimensional bin-packing problem in the context of manufacturing operations is *assembly-line balancing* (Scholl and Voss 1996). In the assembly-line balancing problem, the items are job tasks and the one-dimensional set of weights consists of task times for each job. The goal is to assign tasks to workstations such that the sums of the task times at each workstation are approximately equal. A balanced assignment of the workload enables the assembly line to avoid bottlenecks, thus increasing its output rate and operating efficiency.

One-dimensional bin-packing problems can be formulated using a variety of different objective functions and constraints. In traditional versions of the bin-packing problem, the weight capacity of the bins is fixed, and the goal is to pack items into the weight-constrained bins with the objective of minimizing the total number of bins. Another variation, sometimes referred to as the *modified bin-packing problem* (Brusco et al. 1997; Rao and Iyengar 1994), is modeled using a fixed number of bins with no weight capacity and the objective is to pack items into the bins such that the sums of the item weights within each bin are as evenly distributed as possible. Throughout the remainder of this paper, we focus on this ‘modified’ version of the bin-packing problem because of its relevance to problems in psychology.

There are a variety of possible psychological applications of one-dimensional bin-packing. For example, an analyst might wish to assign a set of students (items) to a collection of groups (bins) such that some measure of within-group diversity or heterogeneity (weights) is balanced across the groups (see, for example, Baker and Powell 2002). Another possible bin-packing application could pertain to the assignment of subjects to treatments based on predictor scores (van der Linden 1981). Perhaps the most prominent psychometric examples, however, occur within the context of test assembly. Consider, for example, the problem of splitting a set of test items¹ to produce two or more parallel test forms² that are roughly equal with respect to total variance (van der Linden 2005, Chap. 5; van der Linden and Boekkooi-Timminga 1988). In this example, the test items are the items, the subtests are the bins, and the item weights are statistical measures derived from item difficulty scores and item discrimination scores.³ We begin with a very general model of the one-dimensional bin-packing

¹Test items are the questions asked in a test; examinees are supposed to select appropriate answers and submit these to be scored and interpreted. Items are the building blocks of any test. They typically consist of an item stem, which contains the stimulus materials to which the examinees are to respond, and a system of response options—a means for the examinees to record their responses.

²Certain testing situations might require us to measure a given attribute using two different tests (i.e., two different sets of items). Tests measuring the same attribute are commonly referred to as alternate forms of a test. If in addition they satisfy certain psychometric properties, then alternate test forms are called parallel. The classical true-score model represents an observed score, Y , as the sum of true-score and error-of-measurement components: $Y = \tau + \varepsilon$. This leads naturally to the notion of parallel measurements, Y and Y^* , scores on tests that yield a shared true score, τ , and errors of measurement, ε and ε^* , assumed to be uncorrelated with the true score and with each other, and to have equal variances.

³In classical test theory, the item difficulty score—or item difficulty for short—is defined as the proportion of examinees in a sample who master an item; hence, an item with a high difficulty is “easy” and one with a low difficulty is “difficult”. Item discrimination is defined as the correlation between the score on a single

problem as a linear zero-one integer programming formulation, with no restrictions on the number of items in each bin. Next, we gravitate toward a one-dimensional bin-packing problem that has particularly relevant applications in psychology. This model includes the incorporation of constraints that require an equal number of items in each bin. Most notably, we focus on an item-to-bin assignment problem, which is a constrained version of one-dimensional bin-packing that uses the popular *minimax* objective function (Adema 1992; van der Linden and Boekkooi-Timminga 1988, 1989). In addition to the splitting of test items, this problem is also relevant to the problem of assigning tasks in a multiprocessor scheduling system with the goal of minimizing makespan (Dell’Amico and Martello 1995). Solution procedures for this one-dimensional minimax bin-packing problem with bin size constraints, which we refer to as “MINIMAX_BSC”, can be classified into two categories: (1) exact procedures that guarantee a globally-optimal solution, and (2) approximate procedures that do not provide a global-optimality guarantee.

One exact solution approach entails the direct solution of mixed zero-one integer linear programming (MZOILP) formulations using *general-purpose* mathematical programming software programs such as CPLEX (ILOG 2009) and Xpress-MP (Dash Optimization 2001–2007). This approach is especially popular in the test assembly literature (van der Linden 1998; Veldkamp 2005), with CPLEX coming highly recommended (van der Linden, 2005, Chap. 4). Although CPLEX typically *finds* feasible solutions to MZOILP formulations of MINIMAX_BSC quickly, our experience is that it is frequently unable to obtain confirmed globally-optimal solutions within a reasonable amount of computation time. We explored the possibility of an alternative exact solution method, a *specific-purpose* branch-and-bound algorithm, which can often produce a guaranteed globally-optimal solution for small problems (50 items or fewer) in far less time than CPLEX. Unfortunately, this specific-purpose branch-and-bound procedure is hampered by scalability limitations, which render the method computationally infeasible for problem instances of practical size (e.g., Veldkamp characterizes typical test assembly problems as consisting of several hundred to several thousand items).

In light of the inherent limitations of the exact solution methods, we developed an approximate procedure for MINIMAX_BSC that is based on the simulated annealing algorithm (Kirkpatrick et al. 1983). Simulated annealing has proved to be quite effective for other classes of one-dimensional bin-packing problems (Brusco et al. 1997; Kämpke 1988; Rao and Iyengar 1994), and has also been successfully deployed in a number of related test assembly applications (van der Linden 2005, Chap. 6; Veldkamp 1999). Our experiments with the proposed simulated annealing heuristic for MINIMAX_BSC demonstrate that it generally provided better solutions than CPLEX in less computation time for problems with up to 6000 items and 300 bins. In light of its experimental performance, as well as its ability to accommodate both linear and nonlinear objective functions, we favor the simulated annealing approach over the exact procedures for MINIMAX_BSC. If an analyst is committed to using commercial mathematical programming software to solve these problems, then we recommend that the simulated annealing heuristic be used to provide exceptionally good initial starting solutions (e.g., as *.MST files in CPLEX).

In Sect. 2, we present several formulations of one-dimensional bin-packing problems, with particular emphasis on MINIMAX_BSC, which is apt to have the greatest importance

item and the total test score. Items with a high discrimination score discriminate well between examinees with high and low ability levels, whereas items with low discrimination scores do not. Item difficulty and discrimination are always estimated during item pretesting, and specify the typical order of magnitude of the estimation errors.

for psychological applications. In Sect. 3, we present the proposed simulated annealing heuristic and demonstrate the algorithm using a small numerical example. A comparison of the simulated annealing heuristic to attempts at exact solution of the MZOILP formulations using CPLEX 12.1 is provided in Sect. 4. The paper concludes with a summary and discussion of extensions in Sect. 5.

2 One-dimensional bin-packing problems

2.1 Mathematical programming formulations—no restrictions on items per bin

We begin by describing a version of the bin-packing problem that is similar to one studied by Rao and Iyengar (1994), and Brusco et al. (1997), who used a squared deviation from target as the objective function. One obvious approach for accomplishing the same goal with a linear objective function is to use the absolute deviation instead of the squared deviation. However, we adopt another approach that is especially popular in the test assembly literature (van der Linden and Boekkooi-Timminga 1988), which is to minimize the maximum item-weight sum across all bins. The model assumes that there are I items with weights of w_i for $1 \leq i \leq I$. The binary decision variables of the bin-packing problem, v_{ib} , assume a value of 1 if item i is assigned to bin b , and 0 otherwise, for $1 \leq i \leq I$, and $1 \leq b \leq B$. There is a single continuous variable, Z , which represents the maximum item-weight sum across all bins. The one-dimensional bin-packing formulation using minimization of the maximum item-weight sum (MINIMAX), which has a single continuous variable, Z , and $I \times B$ binary variables, is as follows:

$$\text{Minimize: } Z \quad (1)$$

$$\text{subject to: } Z \geq \sum_{i=1}^I w_i v_{ib} \quad \forall 1 \leq b \leq B; \quad (2)$$

$$\sum_{b=1}^B v_{ib} = 1 \quad \forall 1 \leq i \leq I, \quad (3)$$

$$v_{ib} \in \{0, 1\} \quad \forall 1 \leq i \leq I, 1 \leq b \leq B. \quad (4)$$

$$Z \geq 0. \quad (5)$$

For the MINIMAX objective function, the resulting formulation is a MZOILP model. The resulting formulation can be tackled using standard commercial mathematical programming software; however, there is no guarantee that a confirmed optimal solution will be obtained. The objective function (1) represents the minimization of Z , and constraint set (2) ensures that Z is at least as large as the item-weight sums for each bin. Constraint set (3) guarantees that each item is assigned to exactly one bin, constraint set (4) places binary restrictions on the assignment variables, and constraint (5) requires nonnegativity of Z .

2.2 Mathematical programming formulations—restrictions on items per bin and assignments

There are a number of additional constraints that can be imposed on one-dimensional bin-packing problems to make them more appropriate for psychological applications. Typically,

these modifications place further restrictions on the assignments of items to bins. For example, whereas the MINIMAX formulation in the previous section permits the number of items in each bin to vary, an analyst might require solutions where the same number of items is assigned to each bin. This type of constraint could be encountered in applications such as the allocation of students to teams or the assignment of subjects to test conditions. The restriction is especially common in applications pertaining to the splitting of a set of items to form parallel tests of equal length. Consider the test splitting problem described by van der Linden and Boekkooi-Timminga (1988), which consists of a two-stage process (see also van der Linden 2005, pp. 131–133). In the first stage, a collection of I items is grouped into T homogeneous sets of size B . The second stage is a one-dimensional bin-packing problem with the additional constraints that B subtests (bins), each containing exactly T items, should be formed by assigning exactly one item from each group to each subtest. Although the second-stage model could be applied directly to the collection of items without prior grouping, the pragmatic role of stage one should not be underestimated. More specifically, stage one ensures that the test versions consist of questions of comparable difficulty. As cautioned by van der Linden and Boekkooi-Timminga, skipping stage one can result in solutions where some subtests might consist of questions that are all of moderate difficulty, while other subtests have a mix of difficult and easy questions.

Adaptation of the one-dimensional minimax bin-packing formulation to make it concordant with the type of problem encountered in the test splitting literature is straightforward. The modified formulation assumes that there are T sets each consisting of B matched items, resulting in a total of $I = T \times B$ items. For each of the B items ($1 \leq r \leq B$) in each of the T sets ($1 \leq t \leq T$), we assume that an item weight, η_{rt} , is available. In the test splitting context, these item weights would typically correspond to measures of each item's difficulty and/or discrimination.

An adaptation of the MINIMAX formulation, which is similar to the model presented by van der Linden and Boekkooi-Timminga (1988), uses the binary decision variables, x_{rtb} , which assume a value of 1 if item r in set t is assigned to bin b , and 0 otherwise, for $1 \leq r \leq B$, $1 \leq t \leq T$, and $1 \leq b \leq B$. The MZOILP formulation for the minimax objective function when incorporating bin size constraints (MINIMAX-BSC), is as follows:

$$\text{Minimize: } Z; \quad (6)$$

$$\text{subject to: } Z \geq \sum_{r=1}^B \sum_{t=1}^T \eta_{rt} x_{rtb} \quad \forall 1 \leq b \leq B; \quad (7)$$

$$\sum_{r=1}^B x_{rtb} = 1 \quad \forall 1 \leq t \leq T, 1 \leq b \leq B; \quad (8)$$

$$\sum_{b=1}^B x_{rtb} = 1 \quad \forall 1 \leq r \leq B, 1 \leq t \leq T; \quad (9)$$

$$Z \geq 0, \quad (10)$$

$$x_{rtb} \in \{0, 1\} \quad \forall 1 \leq r \leq B, 1 \leq t \leq T, 1 \leq b \leq B. \quad (11)$$

The objective function (6) of MINIMAX-BSC aims at minimizing the maximum sum of the weights across all components. A theoretical lower-bound on the objective function is:

$$Z_{LB} = \frac{1}{B} \sum_{r=1}^B \sum_{t=1}^T \eta_{rt}. \quad (12)$$

The objective function is closely linked to constraint set (7), which ensures that Z must be at least as large as the sum of the weights for each of the B bins. Constraint set (8) guarantees that each bin will be assigned exactly one of the items from each of the T matched sets. Similarly, constraint set (9) requires that each of the B items in each of the T matched sets is assigned to exactly one bin. Constraint (10) enforces a non-negativity restriction on the continuous decision variable, Z . Constraint set (11) places the binary restrictions on the assignment variables. The MINIMAX-BSC formulation has $T \times B^2$ zero-one integer variables, 1 continuous variable, and $2I + B$ constraints (not including the nonnegativity and binary restrictions).

As noted by van der Linden and Boekkooi-Timminga (1988), for the case of $B = 2$ (i.e., splitting a set of items into halves), the MINIMAX-BSC model is equivalent to matching the sums of the item weights for each half to the greatest extent possible. Our experience is that the model also tends to produce balanced sums across components for most problems where $B > 2$.

Throughout the remainder of this paper, we will focus our description of methods on the MINIMAX-BSC formulation. There are two reasons for this emphasis. First, the MINIMAX-BSC model is most closely related to previous work in the test splitting literature (van der Linden 2005, Chap. 5; van der Linden and Boekkooi-Timminga 1988) and, accordingly, seems to have the most direct relevance to research and development in psychological testing. Moreover, MINIMAX_BSC can also be applied to other relevant problems in psychology, such as the assignment of subjects to treatment groups. Second, the focus on a single formulation enables continuity of presentation, avoids unnecessary notational clutter, and averts confusion that could stem from switching between objective functions. In the next section, we describe a simulated annealing heuristic for the MINIMAX-BSC formulation.

3 A simulated annealing heuristic

3.1 Details of the heuristic

We developed a simulated annealing heuristic for the one-dimensional bin-packing problem corresponding to the MINIMAX-BSC formulation. For convenience of presentation, we define $\mathbf{a} = [a(i)]$ as a vector with elements $a(i)$ representing the bin to which item i is assigned for $1 \leq i \leq I$, $\eta(i)$ as the objective function coefficient (i.e., the weight) of item i for $1 \leq i \leq I$, and z as the objective function value corresponding to the assignment vector. The algorithm requires an initial feasible assignment of items to bins, $\mathbf{a}^* = [a^*(1), a^*(2), \dots, a^*(I)]$ with objective function value, z^* . In our implementation, we produce an initial solution by randomly assigning each of the B items in matched set t to one of the B bins, such that no two items in the set are assigned to the same bin. This process is performed for all $1 \leq t \leq T$ to obtain a feasible assignment of all I items. The neighborhood search process for generating trial solutions with the simulated annealing heuristic begins with the random selection of one of the t sets, t' . Next, the item, r' , in set t' that is assigned to the first bin, b' , with the maximum sum of item weights is identified. Another item, $r'' \neq r'$, with bin assignment b'' , is randomly selected from t' and the bin locations of

r' and r'' are swapped. The effect of this neighborhood search operation on the objective function is computed as follows:

$$\Delta(r', r'') = \max \left\{ \eta(r'') - \eta(r') + \sum_{i=1}^I (\eta(i) | a(i) = b'), \right. \\ \left. \eta(r') - \eta(r'') + \sum_{i=1}^I (\eta(i) | a(i) = b'') \right\} - z. \quad (13)$$

If $\Delta(r', r'') \leq 0$, then the exchange of the bin locations for the two items is *accepted*. For example, suppose that the sum of weights for bins b' and b'' are 100 and 95, respectively. If $\eta(r'') - \eta(r') = -3$, then swapping the bin locations will yield new item-weight sums for b' and b'' of 97 and 98, respectively. Thus, the swap lowers the maximum sum for bins b' and b'' from 100 to 98 (i.e., $\Delta(r', r'') = -2$), and the exchange of the bin locations should be accepted.

If $\Delta(r', r'') > 0$, then the exchange of the bin locations increases (worsens) the maximum sum, and the acceptance of the new solution is controlled by the magnitude of the worsening, as well as the current temperature (*temp*) of the simulated annealing algorithm. We determine the initial value of *temp* by finding the largest value of $|\eta(r'') - \eta(r')|$ across all feasible exchanges. The temperature is held constant for a fixed number of trial solutions known as the temperature length, *templength*. After every *templength* trial solutions, the temperature is reduced by a cooling parameter, *cool*. The algorithm terminates when a complete cycle through a temperature length fails to result in the acceptance of any trial solutions. A technical description of the simulated annealing heuristic is as follows:

Step 0. Randomly generate an initial assignment of items to bins, $\mathbf{a}^*$ with corresponding minimax objective function value z^* . Set $\mathbf{a} = \mathbf{a}^*$ and $z = z^*$, $tl = 0$ and $flag = 0$. Choose $0 < cool < 1$, *templength*, and obtain an initial temperature, *temp*.

Step 1. Randomly select a set t' . Find the item, r' , in set t' that corresponds to the first bin, b' , that has the maximum sum of item weights. Randomly select a second item, r'' ($r'' \neq r'$) with assignment to bin b'' , from set t' . Compute $\Delta(r', r'')$ using (13).

Step 2. If $\Delta(r', r'') > 0$, then go to Step 4; otherwise, go to Step 3.

Step 3. Set $flag = 1$, $a(r') = b''$ and $a(r'') = b'$. Check the objective function value, z , for the new solution and, if $z < z^*$, then also set $\mathbf{a}^* = \mathbf{a}$ and $z^* = z$. Go to Step 5.

Step 4. Generate a uniform random number, *rnd*, on the interval $[0, 1]$.

If $rnd < \exp(\frac{-\Delta(r', r'')}{temp})$, then set $flag = 1$, $a(r') = b''$ and $a(r'') = b'$. Go to Step 5.

Step 5. If $tl < templength$, then set $tl = tl + 1$ and go to Step 1; otherwise go to Step 6.

Step 6. If $flag = 1$, then set $flag = 0$, $temp = cool \times temp$, $tl = 0$ and return to Step 1; otherwise, STOP.

Step 0 of the algorithm initializes the parameters and installs the incumbent solution $\mathbf{a}$ as the best-found solution, $\mathbf{a}^*$. Step 1 generates the trial solution, which is evaluated in Step 2. It is important to note that trial solutions in Step 3 with $\Delta(r', r'') \leq 0$ must be evaluated to determine if the maximum item-weight sum has been reduced. Even if $\Delta(r', r'') < 0$, this does not guarantee that z was improved because there can be *multiple* bins with the same maximum item-weight sum. Step 4 permits the probabilistic acceptance of trial solutions that worsen the objective function. We observe that both Steps 3 and 4 set $flag = 1$ if a trial solution is accepted. Step 5 passes control to Step 1 if the temperature length has not been achieved; otherwise the control is passed to Step 6. If $flag = 0$ at Step 6, then none

of the *templength* trial solutions were accepted at the current temperature and the algorithm terminates with $\mathbf{a}^*$ as the best-found solution. If $flag = 1$, then the temperature is cooled, the counter for the temperature length is reset to zero, and control is returned to Step 1.

It is worth noting that our proposed simulated annealing heuristic for MINIMAX-BSC uses its own built-in morph list of similar items. This is attributable to the fact that the items are assumed to be pre-processed into matched sets, and the bin location of an item can only be exchanged with a bin location of an item from the same matched set. For problem instances where the items are not pre-processed into matched sets, we recommend forming a morph list for each item that consists of the ω items that are most similar with respect to item weight. Brusco et al. (1997) found that $\omega = 10$ was an effective choice. The neighborhood generation process would then be modified by randomly choosing an item, r' , and then randomly choosing a second item, r'' , from the morph list of r' that is not in the same bin as r' .

The simulated annealing heuristic for MINIMAX-BSC was written in Fortran 90. In addition to this simulated annealing program, we also have written a Fortran 90 implementation of a specific-purpose branch-and-bound algorithm for MINIMAX-BSC, as well as a program for generating the MZOILP. The branch-and-bound algorithm uses standard depth-first search of item assignments, and is comparable in design to procedures described by Klein and Aronson (1991) and Brusco and Stahl (2005) within the context of clustering problems. All MZOILP formulations were solved using CPLEX 12.1 ILOG (2009), and all computational results were obtained using a 2.2 GHz Pentium IV PC with 1 GB of RAM.

3.2 Examples

To illustrate the simulated annealing heuristic and compare its performance to mixed zero-one integer linear programming via CPLEX, we present a small numerical example that is concordant with the test item-to-component assignment problem considered by van der Linden and Boekkooi-Timminga (1988) and van der Linden (2005, Chap. 5). More specifically, we assume the case of I dichotomous items where, for each item $1 \leq i \leq I$, difficulty, π_i , is measured by the percentage of examinees who obtained the correct answer, and discrimination, ρ_i , is measured by split-half reliability.⁴ Further, we assume that these I items have been matched into $T = I/B$ sets of size B , perhaps using the stage one mathematical program described by van der Linden and Boekkooi-Timminga, which can be solved exactly using a network flow algorithm (Armstrong and Jones 1992). Finally, we use van der Linden and Boekkooi-Timminga's approach for establishing the objective function coefficients, which are computed using the following formula:

$$\eta(i) = \phi\pi_i + (1 - \phi)\pi_i(1 - \pi_i)\rho_i \quad \text{for } 1 \leq i \leq I. \quad (14)$$

assuming a weighting constant of $\phi = .5$.

The example consists of $I = 36$ items that have been matched into $T = 12$ sets that each consist of $B = 3$ similar items. The item labels and values of π_i , ρ_i , $\eta(i)$, and the set index

⁴The precision of measurement of a test score is typically referred to as its reliability and is defined in classical test theory as the squared correlation between the observed score, Y , and the true score, τ . The split-half method is one among many techniques for estimating the reliability of a test: a single test of I items (suppose I an even number) is administered once only, and the items are split (in some way) into two subtests, each of $I/2$ items; the correlation between the scores of the two half-tests represents their reliability. The reliability of the total test score on I items can then be approximated by the Spearman-Brown prophecy (or correction) formula.

Table 1 Example 1 test problem and optimal assignment of items to components

Problem input information					Optimal assignments		
Item #	π_i	ρ_i	set (t)	η_i	Component 1	Component 2	Component 3
1	.30	.28	1	.1794000	.1794000		
2	.32	.36	1	.1991680			.1991680
3	.39	.22	1	.2211690		.2211690	
4	.56	.39	2	.3280480		.3280480	
5	.51	.44	2	.3099780	.3099780		
6	.48	.40	2	.2899200			.2899200
7	.59	.53	3	.3591035		.3591035	
8	.61	.54	3	.3692330	.3692330		
9	.67	.63	3	.4046465			.4046465
10	.66	.59	4	.3961980	.3961980		
11	.62	.57	4	.3771460			.3771460
12	.74	.62	4	.4296440		.4296440	
13	.75	.59	5	.4303125	.4303125		
14	.68	.66	5	.4118080		.4118080	
15	.83	.55	5	.4538025			.4538025
16	.77	.40	6	.4204200		.4204200	
17	.73	.40	6	.4044200			.4044200
18	.75	.45	6	.4171875	.4171875		
19	.64	.44	7	.3706880			.3706880
20	.66	.36	7	.3703920		.3703920	
21	.67	.43	7	.3825365	.3825365		
22	.60	.55	8	.3660000	.3660000		
23	.60	.60	8	.3720000			.3720000
24	.51	.48	8	.3149760		.3149760	
25	.41	.52	9	.2678940			.2678940
26	.33	.46	9	.2158530	.2158530		
27	.44	.54	9	.2865280		.2865280	
28	.67	.36	10	.3747980			.3747980
29	.66	.34	10	.3681480		.3681480	
30	.75	.46	10	.4181250	.4181250		
31	.59	.31	11	.3324945			.3324945
32	.63	.27	11	.3464685	.3464685		
33	.58	.22	11	.3167960		.3167960	
34	.36	.31	12	.2157120	.2157120		
35	.35	.22	12	.2000250			.2000250
36	.37	.30	12	.2199650		.2199650	
Final within-component objective value sums					4.0470040	4.0469975	4.0470025

for each item are displayed in the five leftmost columns of Table 1. We devised a specific-purpose branch-and-bound algorithm for obtaining a guaranteed, globally-optimal assignment shown in the three rightmost columns in Table 1. The solution was obtained in approximately 9.8 seconds and the globally-optimal objective function value is $z^* = 4.0470040$.

Next, we solved the MZOILP formulation for the test problem in Table 1 using CPLEX. The software program obtained a solution to the test problem in only .36 seconds; however, the reported objective function for this solution, $z = 4.0472190$, revealed that it was not optimal!⁵

Why did CPLEX fail to obtain the optimal solution to this problem? The reason stems from the fact that CPLEX uses a default optimality tolerance of .01 %. In other words, if CPLEX finds a feasible integer solution with an objective function value that is within .01 % of the current lower bound for the objective function, then the algorithm terminates. For many problems, this tolerance setting is perfectly acceptable; however, for our test problem, the objective coefficients are measured to seven decimal places, and there are many solutions with objective function values that are the same out to at least 4 or 5 decimal places. We modified the default optimality tolerance to .00001 % using the command ‘set mip tolerances mipopt = .0000001’. After this modification, CPLEX obtained the optimal solution to the test problem; however, it required 361.39 seconds to verify optimality.

Three restarts of the simulated annealing heuristic were applied to the test problem in Table 1 using parameters of *cool* = .9 and *templength* = 5000I. The total computation time across the three restarts was 18 seconds, and the objective function values were 4.0470110, 4.0470070, and 4.0470065. Three additional restarts were performed after increasing the cooling factor to *cool* = .975, which increased the total computation time to 85 seconds. However, the slower cooling schedule enabled the optimal objective value of $z^* = 4.0470040$ be obtained for all three restarts.

One of the salient aspects of this example is that the specific-purpose branch-and-bound algorithm can be used gainfully to solve small instances of this type of one-dimensional bin-packing problem. Unfortunately, the program is not scalable for problems of practical size and, therefore, we do not consider this approach further in this paper. A second important observation is that analysts must exercise caution when judging whether or not a solution obtained by general-purpose mathematical programming software is truly optimal. With respect to this issue, it is important to note that integer programming software packages such as CPLEX often find good solutions early in the search process, yet might not find the global optimum within hours or days. Moreover, even when the optimal solution is located, confirming that the solution is a global optimum might not be practical within a reasonable time limit. The third crucial finding associated with the example is that the simulated annealing heuristic appears to be a fast and reliable tool for one-dimensional minimax bin-packing problems. This issue is explored further in the next section.

4 Computational results

As noted in the introduction of this paper, Veldkamp (2005) indicates that most test assembly problems consist of somewhere between several hundred and several thousand items. Our first computational study focuses on item set sizes at the lower end of this range, whereas the second study considers item set sizes at the higher end. For both of the experiments, we attempted to produce test problems with properties comparable to examples observed in the test assembly literature (e.g., van der Linden 2005, Sect. 5.4; van der Linden and

⁵From an applied point of view, the differences in the objective function values observed between MZOILP/CPLEX and our SA implementation might seem of a rather academic nature, with negligible practical consequences. Within an academic perspective, however, they mark the difference when global optimality is set as a goal; and they emphasize the need for careful consideration of such a claim.

Boekkooi-Timminga 1988, p. 207). More specifically, difficulty and discrimination scores were randomly generated for T sets of B matched items, and Eq. (14) was used to compute values for $\eta(i)$. For each of the T sets, the value of π_i for the first item in the set was randomly selected based on a uniform distribution for the interval $[.3, .8]$. Similarly, the value of ρ_i for the first item in each set was randomly generated based on a uniform distribution for the interval $[.25, .60]$. Denoting the difficulty index for the first item as π' , the difficulty indices for the remaining items in the set were obtained from a uniform distribution on the interval $[\pi' - .1, \pi' + .1]$. Likewise, denoting the discrimination index for the first item as ρ' , the discrimination indices for the remaining items in the set were obtained from a uniform distribution on the interval $[\rho' - .1, \rho' + .1]$. The resulting data sets exhibited properties consistent with those observed in the literature, and we deemed this to be preferable to purely random item weights that exhibited no practical structure.

4.1 Computational study #1: Medium-sized item sets

Our first experiment focused on item sets of $I = 300$ and $I = 600$, which are far larger than the example in Sect. 3.2, but much smaller than the item sets considered in Sect. 4.2; hence, we refer to these as medium-sized item sets (Veldkamp 2005). For each number of items, we generated test problems for values of $B = 2, 3, 4, 5, 10, 20, 30$, and 60 , which resulted in a total of 16 test problems. For the largest test problem in the set ($I = 600$ and $B = 60$), the MZOILP formulation has 36,000 binary variables and 1260 constraints. We attempted to solve each of the 16 MZOILP formulations using CPLEX 12.1 with a time limit of 600 seconds (10 minutes). Three restarts of the simulated annealing heuristic were also applied to each test problem using parameter settings of $cool = .9$ and $templength = 5000I$. A time limit of 200 seconds was placed on each restart so that the total time allotted for the simulated annealing heuristic was no more than 600 seconds. Table 2 reports the mean percentage deviation above the objective function lower bound (Z_{LB}) for the MZOILP approach, as well as each restart of the simulated annealing heuristic. Total computation times are also provided.

The results in Table 2 reveal that CPLEX was able to efficiently obtain the globally-optimal solution for the two test problems (#1 and #9) with $B = 2$ bins. For both of these problems, the optimal solution corresponds to a packing of items such that item sums for the two bins are the same. The simulated annealing heuristic also finds the optimal solution for these two problems on all three restarts. For the 14 test problems in Table 2 that correspond to a number of bins that is greater than two (i.e., $B \geq 3$), the simulated annealing approach outperformed the use of CPLEX to solve the MZOILP. For each of these 14 test problems, every restart of the simulated annealing heuristic yielded a better objective function value than we obtained using CPLEX. The simulated annealing heuristic also consumed far less computation time for many of the problems.

The fact that both the MZOILP and simulated annealing approaches provided small departures from the lower bound for most problems has a tendency to mask the marked superiority of the latter approach. Consider, for example, problem #8 ($I = 300$ and $B = 60$), where the MZOILP solution yielded an objective function value that is approximately 1.06 % larger than Z_{LB} . Although this figure seems modest on the surface, it is more than 50 times greater than the percentage deviations of approximately .02 % obtained by each of the three restarts of the simulated annealing heuristic. Even more drastic performance differences were observed for the 600-item problems. For problem #14 ($I = 600$ and $B = 20$), the MZOILP solution provided an objective function value that is approximately .1014 % larger than Z_{LB} . Contrastingly, the second restart of the simulated annealing heuristic obtained an objective

Table 2 Results for medium-sized MINIMAX-BSC problems

Prob #	I	B	Percentage above the lower bound (Z_{LB})				ILP time (seconds)	Total SA time (seconds)
			MZOILP	SA restart 1	SA restart 2	SA restart 3		
#1	300	2	.0000000	.0000000	.0000000	.0000000	1	260
#2	300	3	.0000069	.0000022	.0000038	.0000054	600	246
#3	300	4	.0000793	.0000440	.0000295	.0000668	600	216
#4	300	5	.0004193	.0001675	.0001251	.0001940	600	225
#5	300	10	.0072646	.0008331	.0008933	.0007291	600	241
#6	300	20	.0609114	.0036929	.0029925	.0025256	600	275
#7	300	30	.0798146	.0080641	.0069225	.0076144	600	220
#8	300	60	1.0691920	.0198578	.0215114	.0204090	600	264
#9	600	2	.0000000	.0000000	.0000000	.0000000	1	446
#10	600	3	.0000021	.0000013	.0000000	.0000000	600	494
#11	600	4	.0000083	.0000062	.0000062	.0000042	600	573
#12	600	5	.0001486	.0000252	.0000187	.0000226	600	461
#13	600	10	.0021433	.0002845	.0002736	.0002028	600	518
#14	600	20	.1014411	.0008253	.0009212	.0007082	600	537
#15	600	30	.2024312	.0019030	.0017152	.0021248	600	553
#16	600	60	.8901048	.0046633	.0047353	.0050954	600	587

Note—for instances where the percentage above the lower bound is ‘.0000000’, the global optimum solution was obtained. The values highlighted in bold reflect the best result obtained for the test problem corresponding to the row. The implementation of CPLEX 12.1 was limited to 600 seconds, and the time reported for the SA heuristic in the last column is the total time across all three restarts

function value that was only .0007 % larger than Z_{LB} . Thus, the percentage deviation from the lower bound associated with the MZOILP solution was more than 120 times greater than that of the simulated annealing solution.

4.2 Computational study #2: Large item sets

Our second experiment focused on large test problems with either $I = 3000$ or $I = 6000$ items. For each number of items, we generated test problems for values of $B = 2, 3, 4, 5, 10, 20, 30, 60$, and 300 bins, which resulted in a total of 18 test problems (labeled #17 through #34 to continue with the numbering scheme from Study #1). For the largest test problem in the set ($I = 6000$ and $B = 300$), the MZOILP formulation has 1.8 million binary variables and 12,300 constraints. This problem, as well as several others, exceeded the storage capacity of our program for generating the CPLEX input. We were able to generate the CPLEX input for 10 of the 18 problems (problems #17–22 and #26–29). We attempted to solve each of these 10 MZOILP formulations using CPLEX 12.1 with a time limit of 900 seconds (15 minutes). Three restarts of the simulated annealing heuristic were also applied to each test problem using parameter settings of $cool = .9$ and $templength = 500I$. A time limit of 300 seconds was placed on each restart so that the total time allotted for the simulated annealing heuristic was no more than 900 seconds. Table 3 reports the mean percentage deviation above the lower bound for the objective function (Z_{LB}) for the MZOILP approach, as well as each restart of the simulated annealing heuristic. Total computation times are also provided.

Table 3 Results for large MINIMAX-BSC problems

Prob #	I	B	Percentage above the lower bound (Z_{LB})				MZOILP time in seconds	Total SA time in seconds
			MZOILP	SA restart 1	SA restart 2	SA restart 3		
#17	3000	2	.0000000	.0000000	.0000000	.0000000	16	270
#18	3000	3	.0000126	.0000002	.0000002	.0000000	900	308
#19	3000	4	.0001067	.0000011	.0000011	.0000007	900	252
#20	3000	5	.0003045	.0000029	.0000026	.0000018	900	306
#21	3000	10	.0062057	.0000153	.0000185	.0000143	900	297
#22	3000	20	.0355978	.0000552	.0000388	.0000460	900	279
#23	3000	30	n.a.	.0000510	.0000589	.0000733	n.a.	359
#24	3000	60	n.a.	.0001495	.0001495	.0001371	n.a.	415
#25	3000	300	n.a.	.0014075	.0013765	.0013765	n.a.	516
#26	6000	2	.0000000	.0000000	.0000000	.0000000	23	657
#27	6000	3	.0000042	.0000000	.0000000	.0000000	900	656
#28	6000	4	.0001326	.0000002	.0000001	.0000002	900	614
#29	6000	5	.0001155	.0000006	.0000008	.0000007	900	591
#30	6000	10	n.a.	.0000022	.0000029	.0000019	n.a.	643
#31	6000	20	n.a.	.0000118	.0000082	.0000097	n.a.	688
#32	6000	30	n.a.	.0000172	.0000180	.0000172	n.a.	801
#33	6000	60	n.a.	.0000362	.0000331	.0000440	n.a.	702
#34	6000	300	n.a.	.0003502	.0004185	.0003578	n.a.	900

Note—for instances where the percentage above the lower bound is ‘.0000000’, the global optimum solution was obtained. The values highlighted in bold reflect the best result obtained for the test problem corresponding to the row. The implementation of CPLEX 12.1 was limited to 900 seconds, and the time reported for the SA heuristic in the last column is the total time across all three restarts. The “n.a.” terms in the MZOILP columns represent instances where the storage of the constraint matrix for the formulation exceeded available computer memory and, therefore, results are not available

The results in Table 3 reveal that, once again, CPLEX was able to efficiently obtain the globally-optimal solution for the two test problems (#17 and #26) with $B = 2$ bins. The simulated annealing heuristic also found the optimal solution for these two problems on all three restarts. In light of the efficiency of obtaining exact solutions for these large instances of $B = 2$ test problems, we conjecture that the special case of MINIMAX-BSC might be solvable in polynomial time. Part of our rationale for this conjecture is that the closely-related problem of minimum-diameter partitioning is NP-hard for the case of three or more clusters, but can be solved in polynomial time using for the two-cluster case using a repulsion algorithm developed by Rao (1971).

For the remaining eight problems in Table 3 where the MZOILP formulation input could be generated, the simulated annealing approach outperformed the use of CPLEX to solve the MZOILP. For each of these eight test problems, every restart of the simulated annealing heuristic yielded a better objective function value than we obtained using CPLEX. The total computation time for the simulated annealing heuristic was also appreciably less than the 900 seconds consumed by CPLEX for these eight problems. In some instances, the simulated annealing solutions were much better than those obtained with CPLEX. Consider, for example, problem #22, where the CPLEX solution produced an objective function value that is approximately .0356 % greater than the lower bound. This figure is more than 900

times greater than the percentage deviation for the best solution obtained by the simulated annealing heuristic (restart 2), which has a deviation of only .0000388 %.

Every restart of the simulated annealing heuristic provided a percentage deviation from the lower bound of .002 % or smaller for the 18 test problems in Table 3. This is an important finding because it shows that good solutions to large problems can be obtained in reasonable computation time with modest parameter settings. It is true that a slower cooling schedule (e.g., increasing the cooling factor from .9 to .975) or longer temperature length (e.g., increasing *templength* to 10,000) would likely enable slightly better solutions to be identified and would tend to reduce variation in the objective function value across the restarts. However, these more aggressive parameter settings would also increase computation time substantially. As it stands, the current parameter settings should enable analysts to handle problems with 10,000 or more items while maintaining reasonable computational effort.

5 Conclusions

We have developed a simulated annealing heuristic for a psychologically relevant one-dimensional bin-packing problem and, subsequently, compared its performance to an approach based on mixed zero-one integer linear programming. The simulated annealing heuristic builds on previous implementations by Kämpke (1988), Rao and Iyengar (1994), and, especially, Brusco et al. (1997). There are several noteworthy special features of our heuristic relative to these previous methods. First and foremost, the neighborhood search method in our implementation exploits the special structure of matched sets of items for this particular one-dimensional bin-packing problem. Second, we use problem-specific information to set the initial temperature, which avoids the selection of an arbitrary value that might be too large or too small. Third, our procedure averts excessive computation time by terminating the algorithm if a complete cycle at a particular temperature fails to result in the acceptance of a new incumbent solution.

We compared our simulated annealing algorithm to an implementation of mixed zero-one integer linear programming using CPLEX 12.1. The results of this comparison showed that, under equal computation time restrictions, the simulated annealing heuristic systematically produced better solutions than CPLEX for test problems where the number of bins was three or more. Moreover, the simulated annealing heuristic yielded very small departures from the lower bound for test problems with medium and large item sets. With modest parameter settings, the proposed simulated annealing heuristic is scalable for test assembly problems involving thousands of items (as can occur in practice; Veldkamp 2005).

The simulated annealing approach also affords considerable flexibility. First, although we focused our analyses on the minimax objective function, there is no reason that the simulated annealing approach cannot be adapted for other linear, as well as nonlinear, objective functions. Second, the simulated annealing approach can also be modified to include (or exclude) various problem constraints. For example, the assumption that the number of items per bin must be the same for all bins could be relaxed, perhaps with the addition of constraints to keep the number of items per bin within some range. Third, adaptation of the algorithm for problem instances where not all items are required to have bin assignments is also relatively straightforward. Fourth, an important extension would be the integration of a simulated annealing within the framework of robust optimization. More specifically, given that parameters of the item selection model are estimated with uncertainty, the development of an algorithm that could yield a robust packing would be highly desirable. Fifth, another obvious valuable extension would be the development of procedures for *multidimensional*

bin-packing problems. For example, in the test splitting context, if difficulty and discrimination were treated independently, then a two-dimensional packing problem might be relevant. Consideration of two or more dimensions in the formation of student teams or subject groups might also be of practical merit. We are in the process of refining our simulated annealing procedure for multidimensional minimax bin-packing problems.

Finally, we note that our results are not intended to diminish the role of integer linear programming approaches for bin-packing and related test assembly problems. Although integer linear programming requires an initial software investment and care must be taken with respect to implementation issues such as optimality tolerance, the approach also affords considerable flexibility for handling a host of auxiliary constraints that might be relevant for practical problems. In some instances, these additional constraints can enable guaranteed, globally-optimal solutions to be obtained in reasonable time, which is a benefit that is not realized with simulated annealing and other heuristics. Moreover, although the incorporation of additional constraints (both soft and hard) and objective criteria can be accommodated within the framework of simulated annealing, this can often present a formidable challenge.

It is also important to recognize that heuristic methods such as simulated annealing need not be considered as distinct *alternatives* to integer linear programming because the methods can be used in conjunction with one another. For example, the simulated annealing heuristic could be used to find a very good solution for the optimization problem, which can provide a starting solution for the integer programming software (e.g., using a *.MST file in CPLEX). Accordingly, the integer programming software might be able to improve on this starting solution, or possibly confirm that it is optimal.

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